Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Resource-Efficient GaN-Based Multi-MHz Power Electronics for Next-Generation Charging and Power Supply Systems			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (√), Advanced Mobility (√), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical · Electronics		Mechanical, Materials	
	Medium Category	Small Category	Medium Category	Small Category
	Power Conversion Equipment	Power Supply Units	Automobile/Rail Vehicle	Electrical and Electronic Systems
Concept of Proposed Technology	The proposed technology addresses the development and experimental validation of GaN-based multi-MHz power electronics for charger-relevant applications, with a focus on high efficiency, compact system design, and reduced use of magnetic core materials. At the core of the project is a galvanically isolated DC/DC converter operating in the multi-MHz switching frequency range, with a target frequency of up to 10 MHz and output power levels relevant for fast chargers and auxiliary power supplies. The approach leverages the high switching speed of GaN power transistors to significantly reduce the size of passive components and to enable the use of coreless (air-core) magnetic components. Conventional chargers and power supplies typically operate at switching frequencies below 1 MHz and rely on ferrite-based magnetic cores, which dominate volume, weight, and material usage. By contrast, multi-MHz operation offers the potential for substantially increased power density and reduced reliance on ferrite materials, contributing to more compact and resource-efficient power conversion systems. The project targets a charger-relevant laboratory prototype that integrates power stage, magnetic components, and thermal design, and that is suitable for realistic operating conditions. While the primary focus is on battery chargers, the developed concepts are directly transferable to automotive onboard and auxiliary chargers as well as power supply units for data-center applications, where efficiency and material usage are of increasing importance.			

Rationale for Joint R&D Project	Multi-MHz power conversion at charger-relevant power levels represents a technically challenging and still insufficiently explored domain, particularly when combined with galvanic isolation and coreless magnetic components. Key challenges include switching losses, parasitic effects, electromagnetic interference, and thermal management under realistic operating conditions. Addressing these challenges requires a combination of fundamental experimental investigation and application-oriented system design, which motivates the joint R&D approach between a research institute and an industrial partner. The research institute contributes in-depth expertise in power electronics, high-frequency converter design, and advanced experimental characterization, enabling systematic exploration of technological limits starting from TRL 2. The Korean SME partner complements this with an industrial perspective on components, manufacturability, and system integration, ensuring that design decisions remain aligned with practical constraints and future product requirements. This collaboration reduces technological risk while enabling the development of product-relevant design knowledge that can be directly exploited in subsequent industrial development activities.
Key Words	1. GaN power electronics 2. Multi-MHz switching 3. Coreless magnetics 4. High-efficiency chargers 5. Power density and resource efficiency

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (2)	**End of the Project**	Level (5)
R&D Objectives	The overall objective of the proposed joint R&D project is to develop and experimentally validate GaN-based multi-MHz power electronics for charger-relevant applications, with a strong focus on high efficiency, compactness, and reduced material usage. Starting from TRL 2, the project addresses the currently unvalidated feasibility of operating galvanically isolated DC/DC converters at multi-MHz switching frequencies (≥ 10 MHz) and power levels relevant for fast chargers. The work explicitly targets the transition toward TRL 5, culminating in a laboratory prototype suitable for charger operation. The specific R&D objectives are: 1. Exploration of multi-MHz operation limits To investigate the practical feasibility of GaN-based switching at multi-MHz frequencies under realistic electrical, thermal, and electromagnetic boundary conditions. 2. Development of a high-frequency switching cell and gate drive concept To design and experimentally validate GaN switching cells and ultra-fast gate driver concepts enabling stable operation in the multi-MHz range. 3. Design and characterization of coreless magnetic components To develop air-core inductors and/or transformers suitable for multi-MHz operation and charger-relevant power levels, with particular focus on losses, thermal behavior, and EMI. 4. Implementation of a galvanically isolated DC/DC converter topology To select, design, and realize a converter topology capable of achieving high efficiency while operating at elevated switching frequencies. 5. Demonstration of a charger-relevant laboratory prototype To integrate switching cell, magnetics, control, and thermal design into a functional prototype representative of a fast charger or auxiliary power supply. 6. Derivation of product-relevant design guidelines To systematically identify feasible operating ranges and trade-offs between switching frequency, efficiency, thermal behavior, and system complexity, providing a basis for future industrial product development.			
R&D Target Specifications	The project aims to achieve the following target specifications: • Input voltage range: approximately 325 V – 500 V DC • Output voltage range: 36 V – 48 V DC • Output power: up to 600 W • Switching frequency: multi-MHz operation with a target of ≥ 10 MHz • Efficiency: target ≥ 95 % at nominal operating point • Galvanic isolation: required • Magnetic components: significant reduction or elimination of ferrite cores through air-core implementations • Demonstrator level: laboratory prototype suitable for charger operation These specifications are chosen to ensure direct relevance to fast chargers for battery-powered devices, automotive onboard or auxiliary chargers, and power supply units for data-center applications.			
R&D Plan	The R&D project is structured into five technical phases over a duration of approximately 36 months. **Phase 1:** System definition and conceptual design (Months 1–6) • Definition of application-relevant boundary conditions and performance targets • Review and selection of suitable GaN power devices and packaging concepts • Evaluation and selection of candidate DC/DC converter topologies for multi-MHz operation • Initial analytical and simulation-based assessment of losses, parasitics, and thermal behavior Outputs: • – System specification and target parameters • Selected topology and device concepts **Phase 2:** Multi-MHz switching cell and gate driver development (Months 7–14) • Design of GaN-based switching cells optimized for ultra-fast operation • Development of gate driver concepts suitable for multi-MHz switching			

- PCB layout optimization with focus on parasitic minimization
- Experimental characterization of switching behavior, losses, and thermal performance

Outputs:

- Validated multi-MHz switching cell
- Experimental data on switching limits and loss mechanisms

Phase 3: Coreless magnetic components (Months 15–22)

- Design and simulation of air-core inductors and/or transformers
- Prototype fabrication of magnetic components
- Experimental evaluation of losses, temperature rise, and EMI behavior
- Identification of feasible magnetic design approaches for charger-relevant power levels

Outputs:

- Characterized coreless magnetic components
- Design rules for multi-MHz magnetics

Phase 4: Converter integration and prototype realization (Months 23–30)

- Integration of switching cell, magnetics, control, and thermal design into a DC/DC converter
- Mechanical and thermal integration suitable for laboratory operation
- Construction of a charger-relevant laboratory prototype
- Efficiency, thermal, and functional testing under representative operating conditions

Outputs:

- Functional laboratory prototype
- Verification of system-level performance

Phase 5: Optimization and transferability analysis (Months 31–36)

- Iterative optimization with respect to efficiency and thermal margins
- Assessment of EMI behavior and mitigation strategies
- Analysis of scalability toward automotive and data-center applications
- Consolidation of results into product-relevant design guidelines

Outputs:

- Optimized demonstrator
- Comprehensive design and development guidelines for industrial exploitation

Technical Risk and Mitigation Strategy

The project addresses a high-risk, high-impact technology domain. The main technical risk is whether multi-MHz (≥ 10 MHz) switching at charger-relevant power levels can be practically realized in combination with galvanic isolation and coreless magnetic components. Although GaN transistors enable very high switching speeds, it is unclear whether stable, efficient, and thermally manageable operation at 10 MHz is achievable in a product-oriented converter, due to potential limitations from switching losses, parasitics, EMI, and thermal constraints. If continuous 10 MHz operation proves impractical, this will be treated as a valuable outcome. The project will then identify the practical upper frequency limits and quantify trade-offs between frequency, efficiency, thermal behavior, and system complexity, resulting in product-relevant design guidelines for industrial charger and power supply applications.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute will be responsible for the scientific conception, experimental implementation, and validation of the proposed multi-MHz GaN-based power conversion technology. The core research activities will be carried out by a full-time doctoral researcher, who will be responsible for the detailed design, implementation, experimental investigation, and documentation of the power electronic converter, including switching cells, gate drivers, magnetic components, and system integration. The doctoral research will be closely supervised and technically supported by two senior professors with complementary expertise in power electronics, high-frequency converter design, and system integration. Their active involvement ensures experienced academic oversight, methodological rigor, and sufficient redundancy to address the technical risks associated with multi-MHz operation at charger-relevant power levels. The research institute's responsibilities include in particular: • System-level concept development and selection of suitable converter topologies • Design and experimental validation of GaN-based multi-MHz switching cells and gate driver concepts • Development, simulation, and experimental characterization of coreless magnetic components • Integration of power stage, magnetics, control, and thermal design into a laboratory prototype • Advanced measurement and evaluation of efficiency, thermal behavior, and EMI • Systematic identification of practical operating limits and derivation of product-relevant design guidelines • Scientific documentation and preparation of results for industrial transfer and further development The research institute will provide the required laboratory infrastructure, measurement equipment, and qualified scientific personnel throughout the project duration.
Expected R&R of Korean SME Partner	The Korean SME partner will contribute its industrial expertise and manufacturing-oriented perspective to ensure the relevance and transferability of the developed technology toward future products. The SME partner's main responsibilities include: • Providing industrial input on component selection, packaging concepts, and system integration • Supporting manufacturability, scalability, and cost-related considerations during the design phase • Manufacturing and supplying prototype hardware such as PCBs, power modules, or magnetic components based on jointly defined specifications • Contributing to iterative design optimization based on experimental feedback • Participating in joint evaluation of prototypes with respect to industrial applicability and product constraints • Supporting the definition of development paths toward pilot-scale or product-oriented implementations Through this role, the Korean SME partner gains early access to validated multi-MHz power conversion concepts and design methodologies, while actively shaping the technology toward industrial feasibility and future commercialization.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The project will deliver a validated prototype and design methodology for multi-MHz GaN-based power converters, directly applicable to next-generation chargers and power supplies. Key impacts include: • Demonstration of high-efficiency (≥ 95 %) operation at multi-MHz frequencies • Significant reduction of magnetic core materials, supporting resource-efficient and sustainable power electronics • Increased power density enabling more compact charger designs • Transferable design guidelines for automotive chargers, auxiliary converters, and data-center power supplies For industrial partners, the project reduces technological risk and accelerates future product development in a strategically important field.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization